

$$Q_1 \quad \therefore G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots$$

且  $T=5$ , 有  $G_5=0$

$$\therefore G_4 = R_5 + \gamma G_5 = 2$$

$$G_3 = R_4 + \gamma G_4 = 4$$

$$G_2 = R_3 + \gamma G_3 = 8$$

$$G_1 = R_2 + \gamma G_2 = 6$$

$$G_0 = R_1 + \gamma G_1 = 4$$

$$Q_2 \quad (1) \quad V_\pi(s) = \sum_a \pi(a|s) \cdot Q_\pi(s, a)$$

推导:  $V_\pi(s) = E_\pi[G_t | s_t = s]$

由全期望公式, 在状态  $s$ , 分解动作  $A_t$ ,

$$\text{有 } V_\pi(s) = E_\pi[E_\pi[G_t | s_t = s, A_t = a] | s_t = s]$$

其中  $E_\pi[G_t | s_t = s, A_t = a]$  即为动作  $T$  价值函数  $Q_\pi(s, a)$

$$\therefore V_\pi(s) = E_\pi[Q_\pi(s, a) | s_t = s]$$

$$\therefore V_\pi(s) = \sum_a \pi(a|s) \cdot Q_\pi(s, a)$$

$$(2) \quad Q_\pi(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')]$$

推导:  $Q_\pi(s, a) = E_\pi[G_t | s_t = s, A_t = a]$

$$= E_\pi[R_{t+1} + \gamma G_{t+1} | s_t = s, A_t = a]$$

$$= E_\pi[R_{t+1} | s_t = s, A_t = a] + \gamma E_\pi[G_{t+1} | s_t = s, A_t = a]$$

$$= \sum_{s'} P(s'|s, a) R(s, a, s') + \gamma \sum_{s'} P(s'|s, a) V_\pi(s')$$

$$= \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')]$$

$$(2) \quad V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')]$$

推导:  $V_\pi(s) = \sum_a \pi(a|s) Q_\pi(s, a)$

$$Q_\pi(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')]$$

$$\therefore V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')]$$



$$Q\pi(s,a) = \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma \sum_{a'} \pi(a'|s') Q\pi(s',a')]$$

推导:  $Q\pi(s,a) = \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V\pi(s')]$

$$V\pi(s') = \sum_a \pi(a'|s') Q\pi(s',a')$$

$$\therefore Q\pi(s,a) = \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma \sum_{a'} \pi(a'|s') Q\pi(s',a')]$$

Q3 (1) 
$$\begin{cases} V(s_1) = R(s_1, \text{上}, s_2) + \gamma V(s_2) \\ V(s_2) = R(s_2, \text{上}, s_4) + \gamma V(s_4) \\ V(s_3) = R(s_3, \text{上}, s_4) + \gamma V(s_4) \end{cases} \Rightarrow \begin{cases} V(s_1) = -0.99 \\ V(s_2) = 1 \\ V(s_3) = -1 \end{cases}$$

(2)  $Q\pi(s_2, \text{上}) = [R(s_2, \text{上}, s_4) + \gamma V\pi(s_4)] = 1 + 0.5 \times 1 = 1.5$

$Q\pi(s_3, \text{右}) = [R(s_3, \text{右}, s_4) + \gamma V\pi(s_4)] = 1 + 0.5 \times 1 = 1.5$

$\pi'(s_3) = \arg \max_a \{Q\pi(s_2, \text{上}), Q\pi(s_3, \text{右})\} = \text{右}$

$\therefore$  决策为“右”

(3) 对于  $s_1 \rightarrow s_2 \rightarrow s_4$  路径;

$$s_1 \rightarrow s_2: Q(s_1, \text{右}) \leftarrow Q(s_1, \text{右}) + \alpha [r + \gamma \max_a Q(s_2, a) - Q(s_1, \text{右})]$$

$$\therefore Q(s_1, \text{右}) = 0 + 0.5 \times (0 + 0.99 \times 0.1 - 0) = 0.0495$$

$$s_2 \rightarrow s_4: Q(s_2, \text{上}) = 0.1 + 0.5 \times (1 + 0.99 \times 0 - 0.1) = 0.55$$

策略: 对于  $s_1: Q(s_1, \text{上}) = 0.1 > Q(s_1, \text{右}) = 0.0495$ , “上”

$$s_2: Q(s_2, \text{上}) = 0.55 > Q(s_2, \text{右}) = 0, “上”$$

$$s_3: Q(s_3, \text{上}) = 0.1 > Q(s_3, \text{右}) = 0, “上”$$

